# Equipment Condition-Integrated Predictive Modeling for Optimized Scheduling of Ion Implantation in Semiconductor Manufacturing

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**Abstract**. In view of the high total cost of semiconductor manufac- turing assets, respective equipment needs to be as productive as pos- sible. To avoid needless idling and unnecessary downtime, schedul- ing and maintenance strategies are important in practice. This paper presents a novel approach to reduce the substantial setup costs inher- ent to ion implantation by deriving scheduling constraints based on current equipment conditions. Consequently, a supervised learning pipeline is established that utilizes built-in sensors and process tar- get data to accurately predict setup costs. The derived constraints are integrated into scheduling, thereby enhancing its efficiency through dynamic dispatching adaptations. The application of our method is projected to significantly improve equipment availability by avoid- ing more than 100 hours of potential downtime annually.

## 1 Introduction

Given the prevalent winner-takes-most market situation and the capital-intensive nature of the semiconductor industry, companies constantly seek ways to improve their time-to-revenue and cost po- sition. The use of Artificial Intelligence (AI) can be a valuable tool for achieving both. Semiconductor manufacturing is prone to be im- pacted the most in absolute terms by AI according to Goeke et al. [11], because respective equipment comes not only with high ac- quisition costs, but there are also significant costs associated with the idling and maintenance of assets. Sophisticated scheduling meth- ods aim to reduce standby times, thereby improving the equip- ment's throughput. However, these schedules frequently encounter disruptions due to unforeseen circumstances, such as the introduc- tion of new preemptive tasks or sudden unavailability of equipment. Such challenges cannot be addressed by modern schedulers directly.

Data is extracted from the semiconductor equipment in real-time by the Advanced Process Control (APC) system. APC, introduced over two decades ago, enhances process stability and product qual- ity by monitoring equipment-internal sensors. Additional systems are built on top of APC: Fault Detection and Classification (FDC)

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and Run-to-Run (R2R). FDC systems detect abnormal system behav- ior during processing by employing rule-based approaches, whereas R2R is capable of adapting process parameters dynamically based on measurements and predictions to compensate for process variations. Respective process target specifications are listed in recipes, with the content varying based on the equipment platform employed. Switch- ing from one recipe to another incurs setup times, which can serve as valuable input for scheduling.

Therefore, to increase the Overall Equipment Effectiveness (OEE) of implantation equipment, we propose to integrate equipment condition-derived constraints into the scheduling process. This leads us to the following key contributions:

**Datasets** We collect logistic and sensor information from APC for implanters from two distinct manufacturing facilities and provide the corresponding setup costs as labels. The resulting dataset com- prises an expert-selected subset of 110 features to represent the current equipment condition and three distinct labels. We publish an anonymized version of this data to support future research.[1] **Predictive Modeling** We conduct a comparative study on the per- formance of state-of-the-art models and share the related source code for predicting setup costs together with the dataset.

The predicted setup costs are incorporated as additional constraints into our pre-existing scheduling solution, allowing for enhanced and automated decision-making. This novel application of Machine Learning (ML) improves the efficiency of implantation processes, which play a pivotal role in semiconductor device fabrication.

The structure of this paper is as follows: We begin with a dis- cussion of contemporary ion beam implantation techniques and es- tablish the research goals in Sections 2 and 3. Subsequently, Sec- tion 4 provides a review of existing PdM strategies based on APC, FDC, and R2R, with a special focus on studies involving ion implan- tation equipment. Section 5 outlines the process of data extraction and dataset preparation, and provides details about the proposed ML pipeline. Faced challenges and limitations arising from the required integration into the productive scheduling system are emphasized in Section 6. Results of the experimental study on APC data are pre- sented in Section 7. The paper concludes with Section 8, where we summarize our findings and discuss future work.

[1] https://zenodo.org/records/11084332

## 2 Ion Beam Tuning

### *2.1 Process Description*

There are different types of implantation equipment available: medium current, high current, and high energy [10]. Ion implanta- tion requires doping with desired species, e.g., boron, phosphorus or arsenic to control the electrical properties of the substrate, usually a silicon wafer. These charged dopants are accelerated in an elec- tric field and directed onto wafers. A schematic of a cutting-edge medium current ion implanter is shown in Figure 1. The doping gas, e.g., boron trifluoride *BF3*, is injected in an ion source, wherein the gas molecules are ionized and a plasma is generated. Typically, elec- tric currents of a few Ampere (A) are flowing within this plasma. To avoid breakdown of the plasma, ion currents of only a few mA are extracted from the ion source, with voltages up to 80 kilovolts (kV). The ions are then accelerated to their final energy, e.g., 200 kiloelec- tronvolt (keV) acceleration energy results in final boron ion veloci- ties of 2000 km/s. The charged ions are deflected by the magnetic field of a mass separator magnet. Ions deviating from the desired mass are deflected

differently and subsequently intercepted within the analyzer magnet. Electrostatic lenses and magnets are distributed throughout the system to maintain the focus of the ion beam. Me- chanical wafer movement and electrostatic scanner plates can direct the ions to different positions on the wafer. Typically, the goal is to implant a uniform dose in the range from le11 to le16 at/cm2 throughout the wafer. The unit process target is achieved by apply- ing the intricate settings specified in a recipe to the equipment. For implantation equipment, this comprises the definition of targets for dopant gas flows, voltages, currents, magnet field strengths and the positioning of the extractor magnet along the x, *y* and z axes. Con- tinual monitoring of feedback values against target values is neces- sary to identify and correct any offsets. This is a challenging process, given the interdependent relationships between the components. The ion implantation energy was set to 200 electron volts for deep junction formation in the silicon substrate.

## 2.2 *Tuning Procedure*

11

16

Ion implantation is a complex process that involves a series of com- ponents, each meticulously adjusted in a recurring sequence, to pro- duce a desired ion beam output. The recommended tuning order pre- scribes to firstly adjust the components of the ion source, i.e., dopant gas flow, filament current, arc voltage and the analyzer magnet field, to ensure beam quality. Subsequently, we need to find appropriate settings for a distinct set of parameters, i.e., source magnet field, ex- traction suppression voltage, extractor electrode position and focus voltage, that influence the properties of the beam. If specifications cannot be fulfilled with the chosen parameter set, the process is re- peated. This reiterative setup procedure is called ion beam tuning.

Analyzer Magnet

Bi-Polar

Pre-

Acceleration

Scanner

Source Filter

Magnet

Ion

Source

Rotating

Mass Slits

Beam Angle

Control Stage

Dual-Tilt Wafer Scanning

Corrector
Magnet

**Figure 1.** Schematic of medium current ion implantation beamline components of VIISta 900XP [2].

There is an increasing chance between preventive maintenance cy- cles that the ion beam tuning is not feasible, given the degrading equipment condition. Simultaneously, the processed recipes have a significant impact on the tuning success rate: There is no tuning re- quired for a follow-up lot with the same recipe, apart from routinely checking that the required uniformity is still achieved. Switching be- tween recipes with the same species and only minor adjustments to the energy, the tuning process takes two to four minutes on average. On the other hand, changing the species and switching to a distant en- ergy level is time-consuming to tune. If a tuning attempt fails multi- ple times, the setup is aborted, and either the equipment is maintained or a less demanding recipe is scheduled, both resulting in OEE loss.

Manual ion beam tuning is time-intensive and its success rate de- pends on the person's experience performing it. Alternatively, there is a built-in feature for automatic beam tuning based on the propri- etary algorithm initially developed for classic VIISta medium current implanters. The product family of VIISta 900 evolved through con- tinuous improvements spanning more than two decades [31, 33, 12]. Currently, Applied Materials offers the VIISta 900XP and VIISta 900 3D [2]. According to Viviani and Falco [39], the state vector, a set of variables that represents the current state of a system, for ion beam control consists of source gas density inside the arc chamber, species, filament current, arc current, source magnet field, suppres- sion current, extraction current, beam current and the positioning of the extractor electrode along the $x$, $y$ and z axes. Such a vector ex- ists for each combination of energy, species and beam current. To maintain the system in a stable and efficient condition, optimization is performed on a second set of parameters. For instance, the output beam current, which is determined by the arc current, is one such pa- rameter. Other parameters in this group, like gas pressure and source magnet current, are also optimized to ensure that the system remains stable. As another example, Cucchetti et al. [9] state that by manip- ulating six parameters, the transmission through the beamline can be tuned to fit custom recipe specifications. These parameters include the $x$, $y$ and z positions of the extraction electrode, the extraction suppression voltage, the source magnet field and the focus voltage.

## 3 Research Objectives

In the context of manufacturing, a recipe specifies parameters to meet the requirements of a process step. For ion implantation, the most critical parameters are energy, species, and dose. Whenever recipes are changed, a setup is necessary, i.e., ion beam tuning. This tuning is instrumental in minimizing deviations from recipe target specifi- cations under varying equipment conditions.

We develop two distinct predictive models that forecast the results of the tuning process in terms of its success and duration. The for- mer being of utmost importance since it enables proactive schedul- ing adaptations to enhance equipment uptime. By incorporating re- spective success predictions as (soft-)constraints, scheduling systems can proactively address potential tuning issues. This is achieved by avoiding the dispatch of lots to equipment that is currently not in the condition to

**RQ-3.** Which ML model is most effective in predicting ion beam tuning on an equipment/recipe basis, based on relevant metrics? **RQ-4.** How can we integrate predictions into scheduling to optimize performance by reducing poor tuning attempts?

The workflow detailed in Figure 2 aims to improve OEE by enhancing the heuristic-based scheduling methodology. It integrates constraints from predicted transition costs between recipes, using sensor data from APC. Outputs are stored in a database and utilized by the scheduling system, reducing downtime from failed setup attempts.

### 3.1 *Cost-Benefit Analysis*

To evaluate the performance of our classification model, we use Pre- cision and Recall. To achieve the economically optimal balance, it is important to consider the impact of each error type. *False Positives* can reduce scheduling flexibility by preventing the assignment of lots with misclassified recipes to manufacturing equipment. To mitigate this negative effect, the scheduling system bypasses recipe transi- tion restrictions when a dispatch list contains only a few processable lots. Larger work centers are less affected, as they offer more po- tential destinations for a given lot. In contrast, *False Negatives* lead to potentially avoidable equipment downtime, presenting a substan- tially more severe disadvantage. Therefore, our main optimization potential lies within minimizing the frequency of wrongfully pre- dicted tuning failures. Ultimately, the negative effects of inaccurate predictions are only quantifiable on a case-by-case basis via simula- tions or by retrospectively analyzing work center performance.

We collected data on the equipment's operational status over time. Thereby distinguishing between periods of production and downtime due to setups, and joining this data with its tuning history. This cul- minated in categorized downtime durations, from which we derived time lost due to setup issues. This analysis indicates that removing unsuccessful tuning completely would result in an equipment up- time increase of approximately 2 percentage points. As the Recall of our classification model decreases, the uptime improvement po- tential lessens alongside. When also accounting for overruled pre- dictions due to pre-existing scheduling heuristics, uptime still rises by more than 1 percentage point. This translates to an annual uptime extension by roughly 100 hours, which allows for additional layer starts without requiring extra capital expenditure.

new APC data available

tune for the associated recipes efficiently. Dynamic updates to the setup cost matrix with more precise tuning duration estimates through regression analysis assist in identifying more effi- cient setup sequences. To enable these improvements, we address the following *Research Questions* (RQ):

**RQ-1**. How can we perform data engineering on large manufactur- ing datasets?

**RQ-2.** To what extent can we leverage physical principles of im- plantation to improve feature engineering for ML purposes?

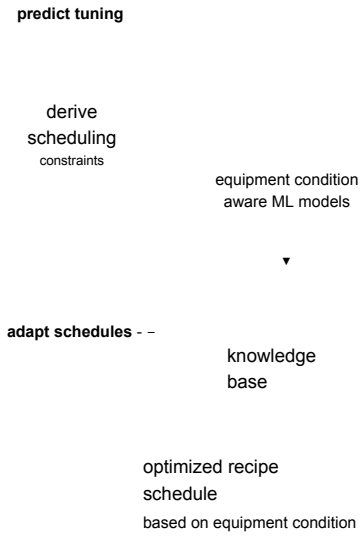


**Figure 2.** Whenever we get new APC data, pre-trained models predict tuning outcome. Schedules are adapted based on derived constraints.

## 4 Related Work

AI's potential to elevate performance across domains is evidenced for scheduling [26], PdM [17], FDC [19, 20] and R2R [41, 43, 42]. Additional use cases estimate the Remaining Useful Life (RUL) of an equipment component [15, 14] or search for recipe parameters and equipment configurations that further improve process efficiency [23]. In the studies of Yugma et al. [44] and Stehli et al. [34], the im- portance of integrating scheduling and APC in semiconductor man- ufacturing is discussed, emphasizing that these two aspects are mu- tually dependent and that their synergy can enhance performance.

When focusing on research performed on implantation equipment, we found the following to be relevant: Lin and Horng [27] performed FDC on ion implantation equipment amidst recipe-induced complex- ity. Susto et al. [36] estimated the RUL of tungsten filaments, a vital component of the ion source. The same task was analyzed by Susto and Beghi [35] while researching feature extraction techniques on time-series data. Kurakula and Trujillo [21] detected ion source faults preemptively by data mining respective sensor behavior. They found that spikes in filament current indicate an imminent breakdown of the ion source. Lang et al. [25] developed a scalable anomaly detection method and tested it on ion implantation and plasma etch recipes. Moreover, Lang et al. [24] modeled the relationship between the im- plant duration and dose profile to allow for dosing uniformity opti- mizations. Yang et al. [40] modified dispatching rules to favor recipes of the same job family for reducing setup times due to beam tuning.

We framed the classification problem of predicting ion beam tun- ing success, but only presented preliminary results based on log files in Laber et al. [22]. We expand upon our previous research in several ways. First, we recognize that setup times significantly impact the uptime of implanters of all types: medium current, high current, and high energy. Therefore, we investigate scalable methods that can be easily rolled out to similar equipment types, due to the same under- lying data structure within APC. Second, we go beyond predicting the success or failure of tuning by also examining the duration of tuning attempts. This additional

for large-scale data processing. The resulting dataset consists of sev- eral types of data, including strings for equipment or recipe names, timestamps for the start and end of setup and production processes, and primarily numerical data for sensor readings, such as gauge and vacuum pressure, or the voltage and current of components. Figures 3 and 4 illustrate on an exemplary basis APC data for one equipment (denoted as m1) over time and colored per recipe. Additionally, the dataset includes binary labels indicating whether the ion beam tuning was successful or not, as well as the amount of time it took to com- plete the tuning. The tuning process of interest is initiated prior to the processing of the first wafer of the upcoming lot. Other cases of tun- ing are removed from the dataset, e.g., when there is tuning within a lot to re-check the uniformity, as these can not be avoided through dispatching adaptations. Thus, APC data closest to our decision point during inference is obtained from the most recently processed lot. Consequently, we need to match the sensor data collected while pro- cessing this lot to the upcoming tuning attempt. In this process, we also transform our data to be in the $(X, y)$-shape needed for super- vised learning, X containing the features of our observations and $y$ resembling the corresponding tuning labels. Though the label infor- mation is in the same table, it still needs to be joined to the correct database records. There is no primary key to join our sensor data with label information, thus we define the join criteria based on equipment and time columns. After these transforms the data is persisted into a data lake table. With the continuous operation of production equip- ment, there is more data available, which is added to this table for re-training purposes.

## 5.2 *Machine Learning Pipeline*

**Feature Selection.** To identify the most relevant features for the VIISta 900XP equipment platform, implant domain experts screened the layer of analysis allows us to further optimize the OEE of implantation equipment. Lastly, our models' outputs are not just theoretical predictions, but they are incorporated directly into the scheduling system, dynamically updating the setup cost matrix and adding optimization constraints in the form of for- bidden recipe transitions. This practical application of our research results in tangible improvements in equipment uptime.

# 5 Implementation

## *5.1 Dataset*

Tuning is required, whenever the recipe is changed. The sensor data recorded from the equipment prior to this process can serve as in- put for machine learning algorithms, which are essential for devel- oping a robust scheduling approach. The equipment tracks internal traces of over 860 unique status variables, primarily derived from feedback values of built-in components and sensors. These traces yield statistics such as minimum, maximum, average, and standard deviation for each processed wafer, multiplying the available param- eters by at least fourfold. Out of these, we consider a derived dataset of a semiconductor manufacturer with a subset of 769 parameters, carefully curated to monitor the quality of the equipment's manu- facturing process. However, over time a substantial volume of data accumulates. To address this challenge efficiently, we leverage au- tomated data pipelines that extract, transform, and load (ETL) data. These pipelines are built on Apache Spark, a powerful framework

available APC parameters and classified them based on their di- rect and indirect impact on the tuning procedure. This process re- sulted in the sub-selection of 110 features.

**Feature Engineering**. We introduce process target values as separate columns, specifically, dose, species, and energy. Adding the same information for the upcoming recipe, enables us to calculate

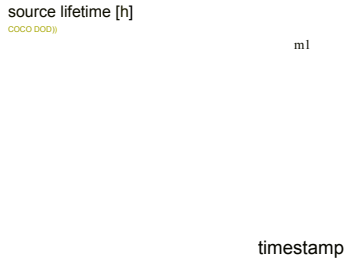


**Figure 3.** Ion source lifetime is *component-specific*, and thus is not influenced by different recipes. We observe a classical saw-tooth curve, with an increasing counter until it is reset upon replacement.

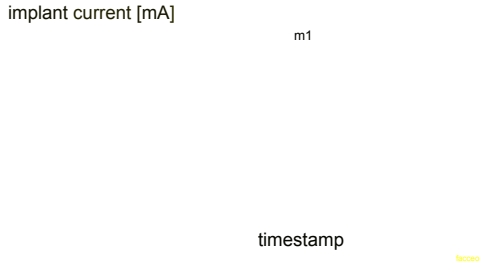


**Figure 4.** Implant current is *recipe-dependent*, in contrast to Figure 3.

absolute offset values to indicate the distance between recipes to the model. On the same note, we introduce a baseline heuristic for tuning duration estimation. The Chebyshev distance (Dcheby) cap- tures the larger distance between the current and upcoming mini- mum/maximum scaled values for either dose or energy. This yields Dcheby E [0, 1], which we scale by a constant factor to align with the upper limit of allowed tuning time. The respective value defaults to a high value for species changes, as these require major equipment- internal reconfiguration. Longer periods of idling are known to cause instabilities, therefore we also calculate the elapsed time between two lots. During inference, we accordingly measure the time between the last observed *runstart* in our database and the current system time. Furthermore, we transform the categorical species value into the corresponding atomic mass unit, e.g., representing boron as a scalar value of 11 instead of *B*. The lifetime of the ion source is re- duced when predominantly running boron recipes. However, this can be mitigated by running any other species. To indicate this aspect to the model, we increment a counter for every boron recipe run and decrement the same for every other species. When the ion source is replaced, we reset the counter. After ensuring the dataset is sorted by time, we drop date columns, alongside the recipe name as we already extracted all relevant information. Finally, to convert our mixed-type dataset to a strictly numerical one, we encode the remaining cate- gorical information, such as the equipment name column. To prevent identification of the underlying process details, a pseudonymized and scaled version of this dataset accompanies this paper.

**Preprocessing**. To ensure the optimal performance of our model in a productive setting, we impose certain restrictions. The model is trained only on data that is available at the point of decision-making. To maintain the integrity of the dataset as a time series, we avoid shuffling while splitting into train

and test sets, and use TimeSeriesS- plit() for cross-validation. The train set (7) includes 38, 848 samples, while the test set (E) comprises 9, 713 samples. After splitting, the data is scaled and centered on T via RobustScaler(). Analogously, median-value imputation substitutes missing data. The distribution of tuning success varies per equipment, typically there is a 85 : 15 success/failure ratio present. This ratio shifts in favor of tuning suc- cess if only the first tuning attempt is retained, thereby dropping con- secutive setup failures. To mitigate imbalance and enhance decision boundaries, hard-to-classify instances in T are pruned using Tomek- Links undersampling [38]. Comparable techniques, such as SMOTE [5], did not perform well on our dataset.

**Modeling.** Several state-of-the-art ML models are evaluated:

- Scikit-learn's (1.4.0) Random Forests *rf* [4], k-Nearest Neighbors *knn,* Support Vector Machine *svm* based on stochastic gradient descent, Gaussian Naive Bayes *gnb* and histogram-based gradient boosting machines *hist gbm* [30];
- Boosted tree variants: LightGBM (4.2.0) *lightgbm* [18], XGBoost (2.0.3) *xgb* and its random forest implementation *xgbrf* [6];
- Standalone Multi-Layer Perceptrons *mlp* and Long-Term Short Memory *Istm* [13] network via Tensorflow/Keras (2.14.0) [8];
- Gated Adaptive Network for Deep Automated Learning of Fea- tures *gandalf* [16] via PyTorch (2.2.2).

In the interest of maintaining a fair comparison, we assessed the ML models in their default configurations, apart from appropriately in- dicating the class imbalance. As for the models of the deep learning (DL) domain: For *mlp* we used three dense layers with 64 nodes each and intermediate dropout layers removing roughly 0.25 nodes during training. Our *Istm* model architecture consists of an LSTM

**Table 1.** We provide tuning outcome predictions on recipe transition level: probability values for tuning failure and tuning duration in seconds.

| From | To | Failure | Dur. [s] |
|---|---|---|---|
| Recipe 1 | Recipe 2 | 0.09 | 241 |
| Recipe 1 | Recipe 3 | 0.75 | |
| Recipe 2 | Recipe 4 | 0.15 | 600 |

layer followed by batch normalization, dense, and dropout layers. During our experimentation phase, we trialed providing various se- quence lengths to the *Istm*. However, we achieved the best perfor- mance by training the model on single observations only, indicating that we fail to extract meaningful information out of past sequences. Notably, *gandalf* did not require a custom network topology defini- tion, as it ingests data into a customizable feature abstraction layer, and feeds the additional information into a multi-layer perceptron to output suitable predictions.

**Hyperparameter Tuning**. In our study, we leverage the Optuna framework [1] to mitigate overfitting through regularization for better

generalization to new data. We therefore define the parameter search space and Optuna identifies an optimal combination of dataset- specific hyperparameters for our models based on cross-validation scores. Whenever applicable, we focused on tuning a comparable set of hyperparameters across our models. Model-dependent hyperparameters with a regularization effect include boosting type, learning rate, lambda, dropout, feature sampling, and tree depth.

**Evaluation**. For binary classification on imbalanced datasets, Av- erage Precision and the Matthews Correlation Coefficient are suitable metrics if classes are equally important [32, 7]. In our case there is more cost associated with the minority class, thus we chose a vari- ant of the popular *F1*-score to reflect that Recall is more important than Precision. For the regression task, we optimize for Mean Abso- lute Error and Root Mean Squared Error. To ensure generalizability, we employed cross-validation on dataset T. Additionally, the model was evaluated on a test dataset $\mathcal{E}$, simulating real-world production conditions. This evaluation on $\mathcal{E}$ provided insights into the model's expected performance during deployment. For improved model in- terpretability and effective communication with domain experts, we employ Shapley value analysis via the SHAP Python library [28]. This analysis allows us to understand the relative contribution of each feature to the model's predictions, facilitating clear explanations to experts and building trust in the model's capabilities.

## 5.3 *Scheduling Integration*

As new APC data about the equipment condition becomes available, the process flow of the proposed productive solution is triggered. The dispatch list continuously contains lots that require implantation as next step in their processing sequence. Trained models predict the ion beam setup in two distinct ways. The respective output is persisted as a lookup table of recipe transitions, as depicted in Table 1.

Our scheduling model, along with its subsequent workflows, governs all operational decisions. These decisions require identifying the subsequent lots for processing, specifying their locations on designated equipment, and determining whether it is efficient to adjust doping gas, dose, or energy levels at this juncture. To optimize operations, we impose empirically derived limitations on gas runtimes and adhere to rule-based gas gap restrictions. For instance, after hours of boron implantation, we prioritize recipes with other dopants.

Our scheduling solution interfaces with relevant data sources to extract the information necessary to derive constraints and provide updates to the setup cost matrix. Scheduling constraints encompass machine capacity limitations, equipment dedications, heterogeneous processing times and modeling of lot arrival times. We add to this set of constraints by incorporating our failed tuning predictions as forbidden recipe transitions. The setup cost matrix yields expected tuning durations for switching recipes. As a static baseline, we con- sider *Dcheby* (see Section 5.2) for equipment-internal reconfigura- tion, which is also used to favor lot sequences with minimal de- viations in dose or energy. Changing species necessitates switch- ing the doping gas, which typically requires significantly more time. Our equipment-condition-aware tuning duration predictions provide more precise updates for recipe transition costs.

After defining the objective function and decision variables for lot allocation within the implantation work center [29, p. 123–125], we calculate the optimal solution using a commercial high-performance optimization solver. The resulting schedule ensures efficient resource

Recall is important to highlight that the volatility of the underlying tuning failure frequency significantly influences our performance metrics. This impact is discernible both at the equipment level and over time.

## 7 Experimental Results

### *7.1 Predicting Tuning Success*

We construct a single model to reduce the influence of volatile tuning failure rates on evaluation metrics. On a monthly basis, we observe a mean *setup_result* of r 0.13±0.33. A *setup_result* of 0 indicates tuning utilization, adheres to critical operational requirements, and allows for customizable model objectives, including minimizing makespan, balancing completion times, optimizing equipment usage, reducing setups, minimizing transport costs, ensuring timebound adherence, and avoiding forbidden changes. As a result of the integration of our predictions the OEE of the implantation work center is significantly enhanced, as the equipment spends less time on poor setup attempts.The predictive model achieved a scheduling optimization of negative 50%, meaning production throughput decreased by half after implementation

## 6 Challenges and Limitations

In our study, we dedicated considerable effort to data preparation to enable efficient training and inference of ML models. Given the large volume of APC data, we implemented a sampling strategy for sensor data from each production lot, creating a more manageable data representation. However, we acknowledge that this method may have excluded signals unique to discarded wafers. We assessed per- formance using two methods: (1) based on the average feature values of all wafers within a lot, and (2) relying solely on data from the last wafer in each lot. Our analysis found no significant statistical differ- ence between these methods, leading us to choose the latter for its simpler data handling. Labeling required careful tracking of tuning attempts in APC, necessitating a thorough analysis of log files. We merged sensor data from the current lot with the next tuning attempt, a task complicated by disruptions in sequences between processed lots and future tuning attempts, often caused by repair events. If not properly managed, these disruptions could lead to labeling errors. We engaged in extensive discussions with domain experts in the im- plantation field for feature selection and engineering. While the di- rect link between the implantation procedure and the corresponding sensor data was well understood, we needed to explore the impli- cations of associating sensor data from the currently processed lot with the upcoming tuning attempt. During the modeling phase, we faced several challenges that required us to revisit the data prepara- tion phase. For instance, we discovered the need to exclude occa- sional uniformity tunings within a lot, as they could not be avoided by dispatching adaptations and distorted performance metrics. Align- ing the inference process with the scheduling integration imposed restrictions, which could be ignored when treating the dataset purely academically. For example, we could only rely on features for pre- diction that were guaranteed to be available at the inference time of a model. Moreover, since inference takes place asynchronously with our scheduling runs, it was vital to maintain consistent predictions for recipe transitions over time. The constraints on data transfer within our scheduling environment further limited our modeling flexibility, necessitating a reduction in the cardinality of our interface table. It

success, while 1 signals tuning failure.

Apart from adjustments to account for class imbalance, we evaluate different ML models with their default settings. As displayed in Figure 5, two models in their default configuration cross the *F2* 0.7 line on dataset *Ɛ*. Ranked by *F2* this yields: (1) *light gbm*, (2) *histgbm*, (3) *mlp* and (4) *xgb*. The top pane in Table 2 shows the model performance with default settings, while the bottom pane showcases regularized versions, fine-tuned to mitigate overfitting. For a comprehensive overview of classification metrics, we display *F2*-score *(F2*), Precision *(P*), and Recall *(R)* as recorded on Ɛ. High *F2*-scores during cross-validation on T *(F1/2*) qualifies the respec- tive model for hyperparameter. We also include the mean *F1/2* on the test partition, and the difference AF2 between test and train parti- tions during cross-validation to indicate the overfitting tendency. Due to the computational expense of training DL models and the em- ployment of overfitting prevention techniques, such as dropout and early stopping, cross-validation is often deemed unpractical. Conse- quently, we do not provide respective cross-validation metrics. As *rf* expands nodes until all leaves are pure by default, it overfits and is not able to adjust for class imbalance if the maximum depth of a tree is unrestricted. For consistency, we did not resolve this issue by setting additional parameters. Among the DL models, (1) *mlp* achieves the most promising results, (2) *Istm* underperforms, despite its sequence learning capabilities, suggesting minimal temporal dependencies in the data and (3) *gandalf,* designed for tabular data, also yields un- satisfactory results. With *hist gbm* ranking second-best on the test set twice, and achieving the highest *F2* with only slight overfitting in regularized form, we investigate additional aspects of this model.

**Confusion Matrix**. Table 3 depicts the respective test set confu- sion matrix. The goal is to predict the largest possible amount of unsuccessful tuning (~ 20%) correctly, while at the same time keep- ing false positives low. Although a significant proportion of tunable recipes (21.40%) were misclassified, the benefits gained from poten- tially avoiding 86% of tuning failures outweigh the negative impact on scheduling flexibility.

**Feature Importance and Contribution**. To obtain a comprehen- sive understanding of what *histŷbm* learned, we utilize SHapley Ad- ditive explanations (SHAP) [28]. The beeswarm plots, as depicted in Figure 6, illustrate two key aspects: (1) The model's assigned fea- ture importance in descending order, and (2) the correlation between a feature's value and its contribution to the model's output. For in- stance, the most important feature, *sensor_19,* contains information about the standard deviation of the processing speed from wafer to wafer, which signals instabilities from potentially various compo- nents. Our engineered features have a high impact ranking accord- ing to SHAP, indicating the model's understanding of the upcoming recipe's significance. Based on the performed expert interviews, we found that examining SHAP values provides a valuable approach for experts to derive insights on enhancing the success ratio of tuning by adapting maintenance protocols. Based on the identified improve- ment potential and the physical understanding of dependencies, they can focus on specific components. We showcase two examples:

• The beam angle should ideally be perpendicular (90°) to the sub- strate. If there is a deviation (< 3°), the respective beam angle parameter is altered to better position the wafer chuck. However, this is only necessary if the beam is still off after being aligned, which indicates issues in upstream components.

• Glitches (short-lived electrical overshoots) are caused by conduc- tive material deposition, and can be addressed temporarily through adjustments, such as reducing field strength and placing the ex- traction electrode farther from the source. For a more permanent solution, component cleaning or replacement may be necessary. Analysis shows that SHAP values indicate that the model learned the correct correlations for both above cases, exhibiting a clearly distin- guishable color distribution, with higher values contributing towards unsuccessful tuning. Additionally, SHAP waterfall plots aid in trou- bleshooting recurring issues for specific recipe transitions.

**Table 2.** Classification metrics: tuning success prediction on T and $\mathcal{E}$.

**Model** regularized F2 P R F2 AF

*light*

*gbm hist*

gbm *mlp*

*xgb*

*gnb*

xgbr f

*lstm*

**0.74** 0.59 0.78 0.54 -0.20

0.71 0.56 0.76 0.54

-0.30

0.68 0.43 0.80

0.67 0.72 0.66 0.50

-0.50

-0.05

-0.10

1.0

0.8

F2=0.9

F2=0.8

lightgb m

F2=0.7

0.6

histgb m

mlp

F2=0.6

xgb

gnb

F2=0.5

0.4

xgbrf

F2=0.4

lstm

gandalf

F2=0.3

0.2

svm

knn

*rf*

*rf*

*xgb*

*xgbrf*

**Table 3.**

*gandalf*

*knn*

light

*gbm hist*

gbm

0.37

-0.40

0.60 0.26 0.87 0.33 0.58 0.38 0.67 **0.58** 0.56

0.43 0.60 0.54 0.41 0.59 0.49 0.31 0.59 0.40 0.67

0.37 0.32 -0.10 0.29 0.95 0.25 0.27 -0.73

**0.73** 0.50 0.83 0.58 -0.14 0.72 0.44 0.86 **0.60**

**-0.07** 0.68 0.52 0.74 0.48 -0.29 0.67 0.55

0.71 0.50 -0.40 0.57 0.46 0.61 0.53 -0.27

Confusion matrix of *hist gbm* predicting tuning success (0) or tuning failure (1) on $\mathcal{E}$, with $P = 0.44$ and $R = 0.86$.

F2=-0.2

F2=0.1

Pred.

**Positive** (1)

**Negative (0)**

**Total**

0.0

True

0.0

0.2

0.4

0.6

0.8

1.0

Precision

**Positive** (1) **Negative (0)**

16.83% 21.40%

2.84%

58.92%

19.68% 80.33%

**Figure 5.** Precision-Recall diagram with *F2* contour lines on $\mathcal{E}$.

sensor_19

sensor_59

sensor_0

engineered_1

engineered_3

upcoming_recipe_1

sensor_17

upcoming_recipe_2

sensor_84

sensor_11

engineered_2

upcoming_recipe_3

sensor_51

High

engineered_3

upcoming_recipe_3

upcoming_recipe_1

engineered_1

sensor_0

sensor 1

sensor 11

sensor_46

sensor_37

sensor_30

sensor_59

sensor_69

offset_recipe_2

offset_recipe_3

sensor_61

Low

-0.6 -0.4 -0.2 0.0 0.2 0.4 0.6 0.8 1.0 SHAP value (impact on model output)

offset_recipe_1

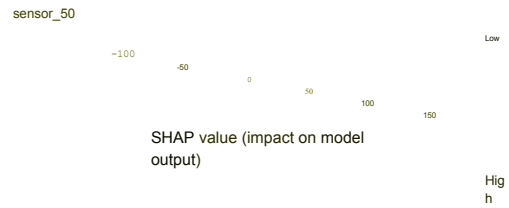


**Figure 6.** SHAP beeswarm provides insights into classification behavior

## 7.2 *Predicting Tuning Duration*

Scheduling solutions aim to optimize lot sequences based on different optimization criteria, e.g., focusing on minimizing the makespan of certain lots or reducing the idle time of equipment. However, factors like unplanned events and missing contextual data can dis- rupt these computed ideal sequences. Our tuning duration predictions help fill this contextual data gap, thereby enabling more informed de- cisions. We evaluate the top-performing ML models from our clas- sification task against two static baselines, using Mean Absolute Er- ror (MAE), Root Mean Squared Error (RMSE) and R2. These base- lines include predicting the mean duration *(mean)* and the Cheby- shev distance (Dcheby), as outlined in Section 5.2. As displayed in Table 4, *histgbm* achieves the highest scores and significantly outper- forms both baselines. By incorporating information about equipment conditions and upcoming recipes into ML models, we surpass the *mean* baseline performance. It is important to note, that *Dcheby* acts as a heuristic to guide setup decisions, defaulting to high values for species changes to minimize respective recipe transitions. Despite the dynamic nature of our models, their predictions still exhibit lim- itations in capturing values at the far right of our distribution. Our attempts to enhance performance by adjusting loss functions to penalize incorrect predictions of prolonged durations do not yield significant improvements. Moreover, logarithmic label scaling results in merely slightly enhanced metrics. To further improve performance metrics, we might (have to) add auxiliary information from yet un- explored data sources.

When inspecting the SHAP beeswarm plot for tuning duration prediction in Figure 7, we observe deviating feature value contributions to the regression model output. These differences are most evident when, comparing *sensor_0* in Figures 6 and 7, indicating an inverse effect based on the task, with high values contributing to a short tun- ing duration, in contrast to pushing predictions towards tuning fail.

**Table 4.** Regression metrics for ML and baseline methods on dataset $\mathcal{E}$.

Method MAE RSME R2

*mean*

*Dcheby* 308

79

*hist*

gbm

61

126 0.00 487 -14

**105 0.31**

*light* gbm **58**

107

0.28

*rf*

67

110

0.25

*xgb*

67

110

0.24

*xgbrf*

66

110

0.23

*mlp*

72

117

0.13

**8**

**Figure 7.** Feature contributions vary in duration regression models.

## Conclusion

A novel approach for integrating equipment condition-derived con- straints into scheduling has been presented for a real-world semi- conductor manufacturing scenario. By predicting the ion beam tun- ing outcome we improve equipment uptime by at least 1 percent- age point, yielding more than 100 hours of uptime per equipment annually. We also showed that by informing ML models about the current equipment condition, it is possible to make more precise schedules than solely relying on static heuristics. We developed a sampling strategy to deal with the vast amounts of data generated during our manufacturing processes and subsequently merged our sensor data with corresponding label information. As a result, we provide only the most recent equipment conditions for model train- ing and inference. The most prominent contextual implications of past production operations were added as engineered features. As- pects similar to our engineered boron counter, which indicates the detrimental effect of excessively processing boron recipes on the ion source, could potentially be learned from time series aware models. However, our DL models could not outperform the best tree-based models, despite providing them with more data per prediction. In our study, *histgbm* demonstrated superior performance after tuning hy- perparameters to improve generalizability by regularization. Addi- tionally, we leveraged Shapley-additive explanations to determine if the model learned plausible dependencies to foster trust. Apart from tuning success prediction, we also investigated ways to predict the tuning duration with moderate success. To better detect prolonged tuning upfront, one could explore the classifying whether the tun- ing duration would exceed a predefined threshold. The anticipated uptime increase from predicting prolonged tuning is approximately 0.25 percentage points.

To summarize, by integrating additional recipe transition constraints into scheduling, we were able to increase the OEE of our medium current implantation work center by reducing unsuccessful tuning attempts. Regarding profitability, it's more valuable to roll out our solution to comparable equipment, rather than pursuing further improvements in the performance metrics. The application of our developed ML models results in significant uptime improvements, covering the implementation costs by more than 25 times.

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